24-3-26

TUTORIAL - 8 (Decision Tree classifier)

ID3 algorithm

Entropy & Information Gain

calculate IG (→ max.) to find best feature (root node)

Entropy ⇒ randomness of data
how data is distributed.

$$-\sum_{i=1}^{N} P_i * \log_2(P_i)$$

Prob. of feature i

$$\log_2 P_i = \frac{\log_{10} P_i}{\log_{10} 2} = \frac{\ln(P_i)}{\ln(2)}$$

IG = entropy(Parent)

$$-\sum_{K} \frac{N_K}{N} \text{entropy}(K) \Rightarrow -\frac{\cdot}{10} \text{entropy}(K) = 0$$

features → K, feature

1) Entropy(Parent or Y)

$$P_{spam} = \frac{5}{10} = 0.5$$

(5 → Total val.)

$$P_{Nspam} = \frac{5}{10} = 0.5$$

$$\text{Entropy} = -\sum_{i=1}^{N} P_i * \log_2(P_i)$$

$$= -\ 0.5 * \log_2(0.5)$$

$$= -(-1)$$

$$= 1$$

Entropy(money keywords) = 0.

↓ they are perfectly classified. so entropy = 0.

1 → spam ✓
↘ Non-spam ✗

0 → N_S ✓
↘ spam ✗

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IG (Money keywords)

= Entropy (Parent)

= 1 - 0 = 1

if sender known is 1 → S → 1 → ~~N~~ Sp
5 → N Spam.

not root node → no. of spam emails =

entropy (Sender known) =

1 → spam (1) = $\frac{1}{6}$ $-\left[\frac{1}{6}\log_2\left(\frac{1}{6}\right) + \frac{5}{6}\log_2\left(\frac{5}{6}\right)\right]$

N. spam (5)

entropy for 1 ⇒

0 → spam = 4

N-spam = 0.

$+\left(0 + \frac{4}{4}\log_2\left(\frac{4}{4}\right)\right)$ ⇒ 0.

entropy for 0

entropy for 0 = 0.

$= -[-0.43 - 0.21]$

$= -[-0.64] \Rightarrow 0.65$

$IG = 1 - \left(\frac{6}{10} \times 0.65 + \frac{4}{10} \times 0\right)$

0.39.

$= 1 - 0.39$

$= 0.61$

not root node. IG < 1.

entropy (has attachments)

0 → spam (2)
N-spam (3)
entropy > 0

1 → spam - 3
N-Spam - 2
entropy > 0.

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entropy(suspiciouswords)

entropy > 0 Min entropy

Max IG

I.G < 1

∴ Money keywords is root node.

if = 1 → Spam

if = 0 → Non-Spam → Leaf nodes

depth = 1 as there is only 1 decision node.

Leaf nodes = 2.

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